



ST BENEDICT'S

SUBJECT	CHEMISTRY FINAL EXAMINATION
GRADE	11
EXAMINER	Mr Ntombela
MODERATOR	Mr Hilder
DATE	27 November 2023
MARKS	165
DURATION	2,5 hours

%

LEARNER NAME: MEMO

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 18 pages, including additional space. A Data Sheet has been provided. Read the questions carefully.
2. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
3. Use the data and formulae sheet whenever necessary.
4. Show your working in all calculations.
5. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
6. Answers must be expressed in decimal format, not left as proper fractions.
7. Where appropriate, express answers to TWO decimal places.
8. It is in your own interest to write legibly and to present your work neatly.
9. A blank page and axis is included at the end of the paper (page 18). If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	Total
Mark									
Moderated Mark									
Question Total	20	30	31	22	19	15	19	9	165

Comment:

QUESTION 1 MULTIPLE CHOICE GR**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	C	D
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Here the option C has been marked as an example.

1.1	A	B	C	D✓
1.2	A✓	B	C	D
1.3	A✓	B	C	D
1.4	A	B	C✓	D
1.5	A	B✓	C	D
1.6	A✓	B	C	D
1.7	A	B	C	D✓
1.8	A	B	C	D✓
1.9	A	B	C✓	D
1.10	A	B	C✓	D

QUESTION 2 CHEMICAL BONDING ZN**30 MARKS**

2.1

2.1.1 Induced dipole (London) forces between its atoms. (1)

Ne✓

2.1.2 Induced dipole (London) forces between its molecules. (1)

N₂✓

2.1.3 A giant covalent network structure. (1)

C✓

2.1.4 Conducts electricity in its liquid phase. (1)

NaCl✓

2.1.5 Ionic bonds. (1)

NaCl✓

2.1.6 Polar covalent bonds. (1)

NH₃✓

2.1.7 Pure covalent bonds. (1)

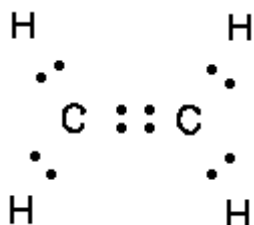
N₂✓

2.2 Carbon and oxygen atoms can bond to form many different compounds.

2.2.1 Define electronegativity. (2)

A measure of the tendency of an atom to attract a bonding pair of electrons✓✓

2.2.2 Draw the Lewis diagram of ethene. (2)



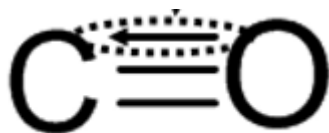
2.2.3 ✓ for any sigma labelled correctly. ✓ pi in double bond. (2)

2.2.4 Define a dative covalent bond. **(IGNORE)** (2)

When a lone pair from one atom overlaps with the empty orbital of another atom✓✓

2.2.5 Draw a Couper diagram of carbon monoxide.

(3)



✓ C and O, ✓ triple bond, ✓ Arrow from O to C

2.2.6 Indicate the polarity of the carbon and oxygen atom in 2.2.5

(2)

✓ positive delta on C, ✓ negative delta on O

2.2.7 With reference to electronegativity, intramolecular forces and molecular shape, determine if CO₂ is a polar or non-polar molecule.

(5)

END = 3,5-2,5 = 1,0>0 ✓ therefore polar bonds ✓

Electron cloud symmetry ✓ as it is linear ✓

Therefore, non-polar molecule ✓

2.3 Study the graph below.

2.4 What phase would Neon be at -200 °C?

(1)

Liquid ✓

A student concludes that as the size of the atoms increases, so does the boiling point of the noble gas.

2.5 With reference to intermolecular forces and the factors influencing their strength, explain the student's conclusion.

(4)

All atoms experience London forces ✓

Number of electrons increases as you move down the group.

The strength of the temporary dipole increases ✓

So the strength of the London force increases (can't just say IMF) ✓ and more energy is needed to overcome the LF. ✓

QUESTION 3

SG

31 MARKS

3.1 Define *catalyst*. (2)

A substance that increases the rate of the reaction✓ but remains unchanged at the end of the reaction✓

3.2 Explain how the data for this experiment would have been collected under laboratory conditions. (3)

Using downward displacement of water✓.

Gas jar is filled with water and inverted✓ over the beehive shelf so that the water is displaced downwards and oxygen is collected upwards. ✓

3.3 Based on their findings, which of the catalysts would have produced the most foam in the quickest time? (1)

MnO₂✓

3.4 Define *rate of reaction*. (2)

The change in amount or concentration per unit time of either a reactant or product✓✓.

3.5 Using the graph, calculate the average rate of the reaction for the catalyst identified in 3.3 during time 0 – 4 ms. Include appropriate units. **COE FROM 3.3** (3)

$$\text{Reaction rate} = \frac{n}{t}$$

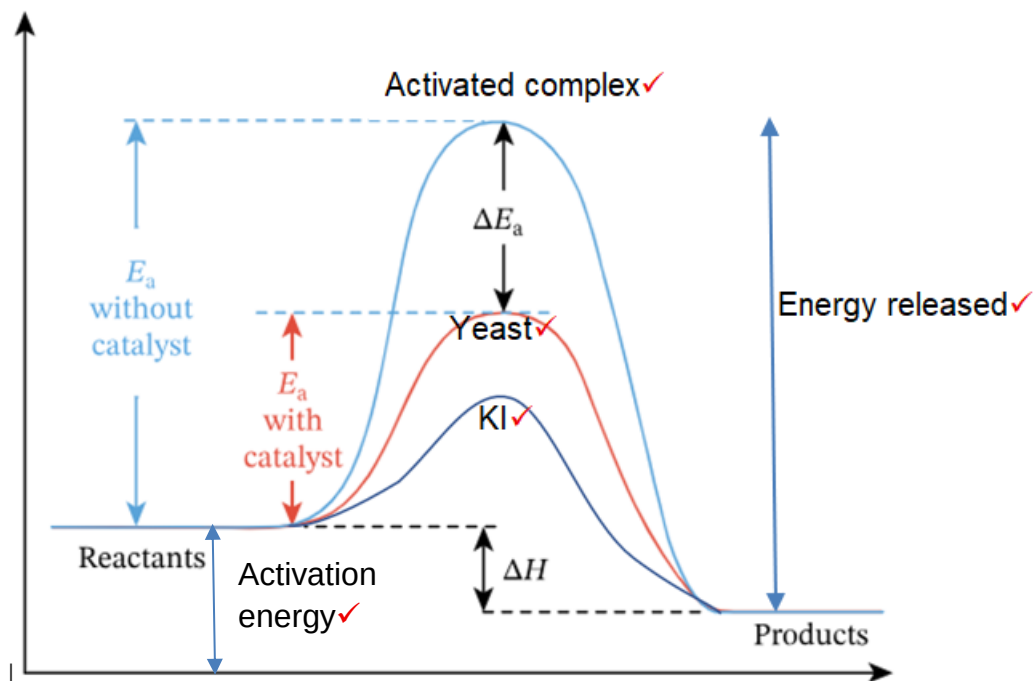
$$\text{Reaction rate} = \frac{40 \times 10^{-3}}{4 \times 10^{-3}} \checkmark$$

$$\text{Reaction rate} = 10 \checkmark \text{ mol} \cdot \text{s}^{-1} \checkmark$$

3.6 Matthew says that the reaction is exothermic because energy is only being released. Joseph, wisely, isn't convinced. Explain fully how Matthew's statement needs to be corrected by referring to the breaking and forming of chemical bonds. (3)

Chemical potential energy is also absorbed✓ the break old the bonds. Chemical potential energy is then released while new bonds✓ are forming in a form of thermal energy✓.

- 3.7 Complete and fully label the energy profile graph below for this reaction. On the same graph, clearly indicate the effect of the addition of yeast in dashed lines and the effect of the addition of KI in solid line. (5)



- 3.8 How would the presence of the catalyst affect the heat of reaction? Explain your answer fully. (4)

No change✓. Catalyst on lowers the activation energy✓.
This provides alternative pathway✓ and does not affect the final energy of the products formed✓

- 3.9 Fully explain (using bullet points) the effect of an increased concentration of hydrogen peroxide on the reaction by referring to collision theory. (4)

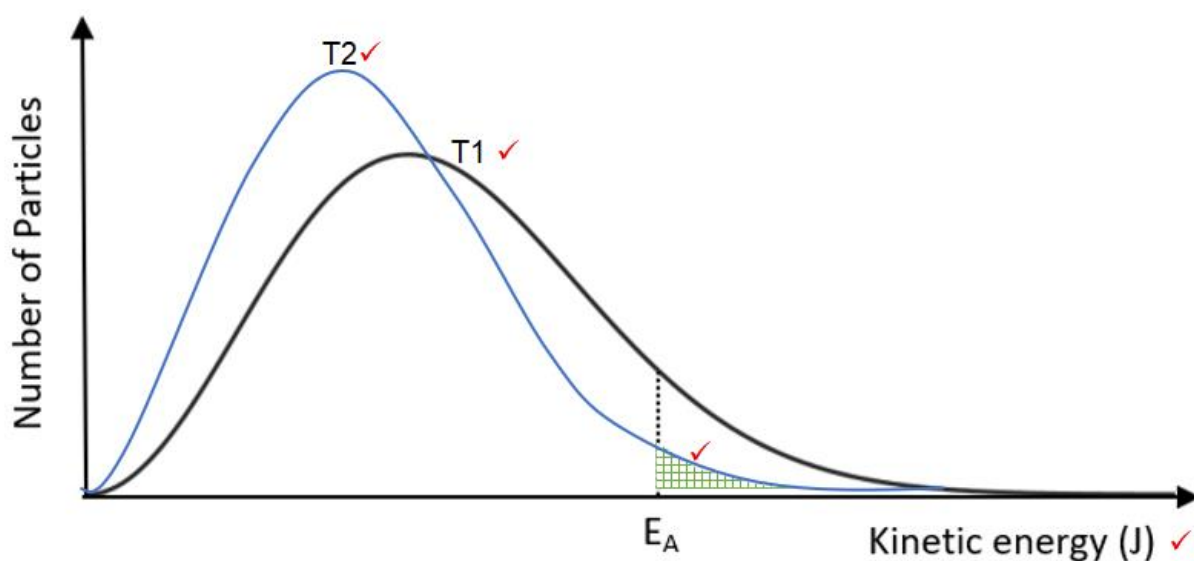
An increase in concentration means that there are more particles per unit volume. ✓

There will be more collisions per unit time ✓.

There will be more effective collisions (with correct orientation and kinetic energy greater than activation energy) per unit time ✓.

The reaction rate increases. ✓

- 3.10 Since both aspiring scientists agree that a temperature change will have an effect on the reaction, draw the corresponding Maxwell-Boltzmann curves showing T1 as the original experiment and T2 as the experiment repeated at a lower temperature. (4)



QUESTION 4 ORGANIC CHEMISTRY FMC**22 MARKS**

4.1 Write down the IUPAC name for compound A. (4)

3✓-bromo✓-2,4✓-dimethylhexane✓

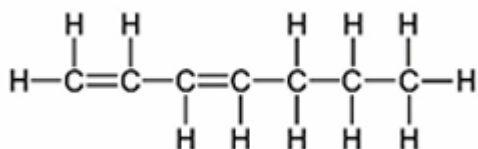
4.2 Draw the condensed-structural formula for compound A. (2)

 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3) \text{✓} \text{CHBrCH}(\text{CH}_3)\text{CH}_3 \text{✓}$ or $\text{CH}_3\text{CH}(\text{CH}_3)\text{CHBrCH}(\text{CH}_3) \text{✓} \text{CHBrCH}(\text{CH}_3)\text{CH}_2\text{CH}_3 \text{✓}$

4.3 NAME the homologous series to which compound B belongs. (1)

Alkenes

4.4 Draw the structural formula for the unbranched CHAIN isomer of compound B. (2)



Correct functional group✓

Correct number of
carbons✓

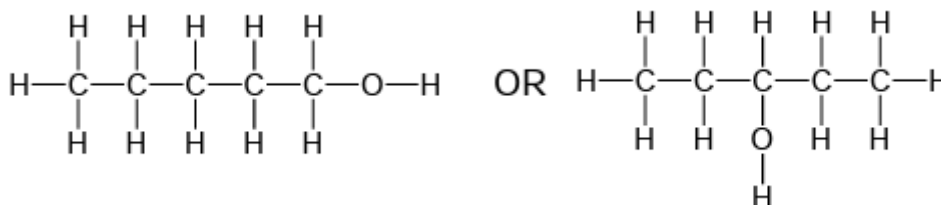
4.5 NAME the functional group present in compound C. (1)

Hydroxyl group✓

4.6 State what is meant by the term positional isomers. (2)

Molecules with the same molecular formula✓ but different positions of their same functional group OR same substituents✓.

4.7 Draw the structural formula for a POSITIONAL isomer of compound C. (2)



Correct functional group✓

Correct number of
carbons✓

- 4.8 Compound C has a significantly higher viscosity than compound B. Account for this difference with reference to the relevant intermolecular forces in each of these compounds. (4)

Compound B has London forces (only) ✓

Compound C has hydrogen bonding ✓

Hydrogen bonds are stronger than London forces (OR compound C has stronger intermolecular forces) ✓

It is more difficult for the particles in compound C to flow past one another resulting in compound C having a higher viscosity ✓

- 4.9 Using molecular formulae, write a balanced chemical equation for the complete combustion of compound B. (4)



QUESTION 5 ORGANIC CHEMISTRY OS**19 MARKS**5.1 Define *hydrocarbon*. (2)

A compound/molecule containing only carbon and hydrogen atoms. ✓✓

5.2 Write down the IUPAC name of compound Y. (2)

Pent✓-1-ene✓

5.3 Classify compound Y as SATURATED or UNSATURATED. (1)

Unsaturated✓

5.4 Write down the IUPAC name for hydrocarbon W. **Pentane**✓✓ (2)

5.5 Identify the general TYPE of reaction for each of the following:

5.5.1 Reaction 1 (2)

Substitution ✓

5.5.2 Reaction 2 (2)

Elimination ✓

5.5.3 Reaction 5 (2)

Addition✓

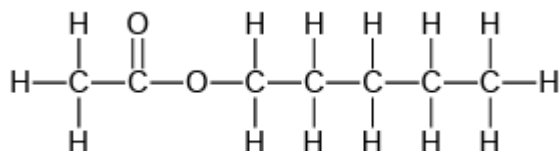
5.6 NAME the specific type of addition reaction represented by reaction 3. (1)

Hydrohalogenation ✓

5.7 Consider reactions 5, 6 and 7 and write down the IUPAC NAME of compound Z. (2)

Pentan-1✓-ol✓

5.8 Draw the structural formula of pentyl ethanoate. (2)



Correct functional group✓

Correct number of
carbons✓

5.9 Write down the NUMBER of the reaction that would take place if the relevant organic compound were heated strongly in the presence of concentrated alcoholic potassium hydroxide. (1)

2✓

QUESTION 6

KH

15 MARKS

- 6.1 Write down the FORMULA of the substance that causes the fizz when the antacid dissolves in water. (1)

CO_2 ✓

- 6.2 Define 1 mole of a substance. (1)

The amount of substance having the same number of particles. ✓ (Not a definition, mark leniently)

- 6.3 Calculate the number of moles of NaHCO_3 in the antacid. (3)

$$n = \frac{m}{M} \quad \checkmark = \quad n = \frac{3,36}{84} \quad \checkmark = 0,04 \text{ mol} \quad \checkmark$$

- 6.4 Determine, using calculations, which substance is the limiting reagent. (4)

$$n(\text{H}_3\text{C}_6\text{H}_5\text{O}_7) = \frac{m}{M} = \quad n = \frac{1,8}{192} = 0,01 \text{ mol Available} \quad \checkmark$$

$$n(\text{NaHCO}_3) = 3 \times 0,01$$

$$= 0,03 \text{ mol Needed} \quad \checkmark$$

$$n(\text{H}_3\text{C}_6\text{H}_5\text{O}_7) < n(\text{NaHCO}_3) \quad \checkmark$$

$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$ is limiting ✓

Accept alternatives methods

- 6.5 Calculate the mass of the reactant in excess. (3)

$$n(\text{NaHCO}_3) = 0,04 - 0,03 \quad \checkmark = 0,01 \text{ mol} \quad \checkmark$$

$$m = 0,01 \times 84 = 0,84 \text{ g} \quad \checkmark$$

- 6.6 Calculate the mass decrease of the beaker contents on completion of the reaction. (3)

$$m = nM \quad \checkmark = 0,03 \times 44 \quad \checkmark = 1,32 \text{ g} \quad \checkmark$$

QUESTION 7

KH

19 MARKS

7.1

7.1.1 Define the term *molar mass*. (2)

Molar mass is defined as the mass (in grams) of one mole of a particular ✓✓

7.1.2 Calculate the molar mass of a salicylic acid molecule (4)

$$M = (7 \times 12) \checkmark + 6 \checkmark + (3 \times 16) \checkmark = 138 \text{ g.mol}^{-1} \checkmark$$

7.1.3 Calculate the mass of aspirin that should have been produced by 3,86 g of the salicylic acid mass produced. (5)

$$n = m / M = 3.86 / 138 \text{ (coe)} \checkmark = 0.028 \text{ moles} \checkmark$$

1 salicylic acid : 1 aspirin therefore 0.028 moles aspirin produced ✓

$$m = n \times M = 0.028 \times (180) \checkmark = 5.04 \text{ g} \checkmark$$

In reality, only 4,65 g of aspirin was produced.

7.1.4 Calculate the percentage yield of aspirin. (3)

$$\% \text{ yield} = 4.65 \text{ g} / 5.03 \text{ g (coe)} \times 100 \checkmark \checkmark = 92.26\% \checkmark$$

7.2 A single aspirin tablet contains 300 mg of aspirin. Calculate the number of hydrogen atoms in the 300 mg tablet. (5)

$$n = m / M = (300 \times 10^{-3}) / 180 \checkmark \text{ (coe)} = 0.00167 \text{ moles} \checkmark$$

1 mole aspirin = 8 mole hydrogens ✓

Therefore $0.00167 \times 8 = 0.0133$ moles H atoms ✓

$$0.0133 \times (6.02 \times 10^{23}) = 8.03 \times 10^{21} \text{ atoms of hydrogen} \checkmark$$

QUESTION 8

FMC

9 MARKS

8.1 Define the term *solvent*.

(1)

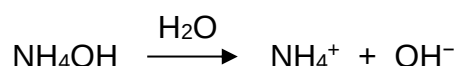
the substance in which another substance is dissolved, forming a solution ✓

8.2 Identify the solvent used in the above solutions.

(1)

water or H₂O ✓

The balanced dissociation reaction for ammonium hydroxide is given as:



8.3 Write balanced chemical equation showing the dissociation of ammonium carbonate.

(3)



8.4 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution.

(4)

$$\text{NH}_4\text{OH}: n = c \cdot V = 0,2 \times 0,25 = 0,05 \text{ mol} \quad \checkmark$$

$$(\text{NH}_4)_2\text{CO}_3: n = c \cdot V = 0,3 \times 0,5 = 0,15 \text{ mol} \quad \times 2 = 0,3 \text{ mol} \quad \checkmark$$

$$c = \frac{n}{V}$$

$$c = \frac{0,05 + 0,3}{0,25 + 0,50} \quad \checkmark$$

$$c = 0,47 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

END OF ASSESSMENT

ADDITIONAL SPACE

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED

[illegible]

[illegible]